

Equity Operational Burden and the Cross Section of Stock Returns

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Abstract

This paper studies the asset pricing implications of Equity Operational Burden (EOB), and its robustness in predicting returns in the cross-section of equities using the protocol proposed by [Novy-Marx and Velikov \(2023\)](#). A value-weighted long/short trading strategy based on EOB achieves an annualized gross (net) Sharpe ratio of 0.32 (0.31), and monthly average abnormal gross (net) return relative to the [Fama and French \(2015\)](#) five-factor model plus a momentum factor of 36 (29) bps/month with a t-statistic of 3.97 (3.27), respectively. Its gross monthly alpha relative to these six factors plus the six most closely related strategies from the factor zoo (Volume to market equity, Days with zero trades, Days with zero trades, Days with zero trades, Tail risk beta, Predicted Analyst forecast error) is 29 bps/month with a t-statistic of 3.03.

1 Introduction

Financial markets process vast amounts of information daily, yet mounting evidence suggests that prices do not instantly reflect all available information. This deviation from the strong-form efficient market hypothesis creates predictable patterns in asset returns that persist across time and markets. While traditional finance theory assumes immediate price adjustment to new information, behavioral and institutional frictions prevent instantaneous information incorporation, leading to gradual price discovery and return predictability.

We examine a novel predictor of cross-sectional stock returns: Equity Operational Burden (EOB), which captures firms’ operational inefficiencies relative to their equity base. Despite the fundamental importance of operational efficiency for firm value, the market’s gradual recognition of operational burden signals creates exploitable return patterns. This paper documents that EOB strongly predicts future returns, with the effect concentrated among stocks where information diffusion is slowest—small firms with limited analyst coverage and low institutional ownership.

The slow diffusion of information provides the economic mechanism linking EOB to future returns. When firms exhibit high operational burden relative to their equity base, this information disperses gradually through the market due to limited investor attention and processing constraints [Hirshleifer and Teoh \(2003\)](#); [Peng and Xiong \(2006\)](#). The complexity of operational metrics and their interaction with equity valuations requires sophisticated analysis that many investors lack the resources or incentives to conduct immediately, creating systematic underreaction to operational burden signals [Hong and Stein \(1999\)](#).

Investor inattention amplifies this gradual information incorporation process. As [DellaVigna and Pollet \(2009\)](#) demonstrate, investors exhibit limited attention to value-relevant information, particularly for complex accounting signals that require careful interpretation. EOB represents precisely such a signal—it demands under-

standing both operational efficiency metrics and their implications for equity valuation. The cognitive costs of processing this information exceed the benefits for many marginal investors, especially in stocks with low visibility and limited intermediation [Merton \(1987\)](#). This creates persistent mispricing that corrects slowly as information gradually reaches the broader investor base.

The concentration of EOB’s predictive power among neglected stocks further supports the information diffusion hypothesis. Small stocks with low analyst coverage experience slower information transmission due to reduced intermediation and higher search costs [Hong et al. \(2000\)](#). These frictions prevent arbitrageurs from immediately eliminating mispricing, as the costs of discovering and trading on operational burden signals exceed potential profits in many cases [Gromb and Vayanos \(2010\)](#). Consequently, we predict that EOB’s return predictability concentrates where attention constraints bind most tightly—among small, illiquid stocks with limited analyst following and low institutional ownership.

Our empirical analysis reveals that EOB strongly predicts cross-sectional returns. A value-weighted long/short trading strategy based on EOB achieves an annualized gross Sharpe ratio of 0.32 and a monthly average return of 29 basis points with a t-statistic of 2.52. The strategy’s alpha relative to the Fama-French six-factor model equals 36 basis points per month with a t-statistic of 3.97, demonstrating that EOB captures distinct return variation beyond established risk factors. After accounting for transaction costs using high-frequency bid-ask spreads, the strategy maintains a net Sharpe ratio of 0.31 and delivers a six-factor alpha of 29 basis points monthly with a t-statistic of 3.27.

The economic magnitude of EOB’s predictive power remains substantial across alternative portfolio constructions. Using NYSE breakpoints with value-weighting, the high-minus-low EOB portfolio generates alphas ranging from 10 to 36 basis points monthly across standard factor models. The strategy loads positively on the market

factor ($\beta = 0.18$) and small-minus-big factor ($\beta = 0.24$), while exhibiting negative exposures to value, profitability, and investment factors. These loadings align with EOB capturing operational inefficiencies that the market gradually recognizes and prices.

Consistent with information diffusion driving the anomaly, EOB’s predictive power concentrates among stocks with higher information frictions. Within the largest size quintile—stocks above the 80th NYSE percentile—the EOB strategy still generates 21 basis points monthly with a six-factor alpha of 35 basis points ($t\text{-statistic} = 3.25$). However, the effect strengthens considerably among smaller stocks where information constraints bind more tightly. The strategy achieves its highest returns of 44 basis points monthly in the smallest size quintile, where limited attention and higher search costs impede information incorporation. This size-contingent pattern directly supports our hypothesis that EOB’s predictability stems from gradual information diffusion rather than risk-based explanations.

This paper contributes to three distinct literatures in asset pricing and market efficiency. First, we extend the growing evidence on gradual information diffusion by identifying a novel operational efficiency signal that predicts returns through underreaction channels. While [Hong et al. \(2000\)](#) document momentum profits arising from slow information diffusion and [Cohen and Frazzini \(2008\)](#) show predictable returns from customer-supplier links, we demonstrate that operational burden relative to equity creates similar gradual price adjustment patterns. Our results complement [Richardson et al. \(2005\)](#) who examine accrual-based predictors, by showing that real operational inefficiencies generate comparable return predictability through information friction channels.

Methodologically, we implement the comprehensive testing framework of [Novy-Marx and Velikov \(2023\)](#) to establish EOB’s robustness across specifications and its incremental contribution to the factor zoo. Unlike many proposed anomalies that

fail out-of-sample or after transaction costs, EOB maintains statistical and economic significance across alternative constructions and after accounting for trading costs. The strategy’s gross monthly alpha relative to six established factors plus six closely-related anomalies equals 29 basis points with a t-statistic of 3.03, confirming that EOB captures distinct return variation. This rigorous approach addresses concerns raised by [McLean and Pontiff \(2016\)](#) regarding data mining and [Hou et al. \(2020\)](#) about spurious anomalies lacking economic foundations.

Our findings carry important implications for market efficiency and investment practice. The persistence of EOB’s predictive power, even among large-cap stocks with substantial analyst coverage, challenges strong-form market efficiency and suggests that operational complexity creates exploitable information asymmetries. For practitioners, EOB offers an implementable strategy that survives transaction costs and enhances mean-variance efficiency beyond existing factors. More broadly, our results underscore how the interaction between firm fundamentals and market frictions generates predictable return patterns, advancing our understanding of price formation in informationally complex markets where attention constraints prevent immediate price discovery.

2 Data

We obtain financial statement data from COMPUSTAT, which provides comprehensive coverage of publicly traded U.S. companies. Our sample includes all firms with available data in the COMPUSTAT annual files, spanning several decades of financial reporting. This database offers standardized accounting information that enables consistent measurement of our variables across firms and time periods. signal construction relies on two key accounting variables. Cost of goods sold (COGS) represents the direct costs attributable to the production of goods sold by a com-

pany during the fiscal year, including raw materials, direct labor, and manufacturing overhead. This expense item appears on the income statement and directly relates to the firm’s operational activities. Common stock (CSTK) represents the par or stated value of common shares outstanding as reported on the balance sheet, reflecting the nominal equity capital contributed by shareholders at issuance. construct the Equity Operational Burden signal by dividing cost of goods sold by common stock (COGS/CSTK). This ratio captures the operational cost intensity relative to the firm’s nominal equity base, providing a measure of how much operational expense burden the firm bears per dollar of stated common equity. We calculate this signal annually using fiscal year-end values from each firm’s financial statements. To ensure data quality, we require non-missing values for both COGS and CSTK, and exclude observations where CSTK equals zero to avoid undefined ratios. The resulting measure provides a standardized metric that allows for cross-sectional comparisons of operational burden across firms with different scales of operations.

3 Signal diagnostics

Figure 1 plots descriptive statistics for the EOB signal. Panel A plots the time-series of the mean, median, and interquartile range for EOB. On average, the cross-sectional mean (median) EOB is 4277.12 (189.75) over the 1963 to 2024 sample, where the starting date is determined by the availability of the input EOB data. The signal’s interquartile range spans 0.85 to 5360.00. Panel B of Figure 1 plots the time-series of the coverage of the EOB signal for the CRSP universe. On average, the EOB signal is available for 6.92% of CRSP names, which on average make up 7.81% of total market capitalization.

4 Does EOB predict returns?

Table 1 reports the performance of portfolios constructed using a value-weighted, quintile sort on EOB using NYSE breaks. The first two lines of Panel A report monthly average excess returns for each of the five portfolios and for the long/short portfolio that buys the high EOB portfolio and sells the low EOB portfolio. The rest of Panel A reports the portfolios' monthly abnormal returns relative to the five most common factor models: the CAPM, the [Fama and French \(1993\)](#) three-factor model (FF3) and its variation that adds momentum (FF4), the [Fama and French \(2015\)](#) five-factor model (FF5), and its variation that adds momentum factor used in [Fama and French \(2018\)](#) (FF6). The table shows that the long/short EOB strategy earns an average return of 0.29% per month with a t-statistic of 2.52. The annualized Sharpe ratio of the strategy is 0.32. The alphas range from 0.10% to 0.36% per month and have t-statistics exceeding 0.96 everywhere. The lowest alpha is with respect to the CAPM factor model.

Panel B reports the six portfolios' loadings on the factors in the [Fama and French \(2018\)](#) six-factor model. The long/short strategy's most significant loading is 0.18, with a t-statistic of 8.16 on the MKT factor. Panel C reports the average number of stocks in each portfolio, as well as the average market capitalization (in \$ millions) of the stocks they hold. In an average month, the five portfolios have at least 629 stocks and an average market capitalization of at least \$1,576 million.

Table 2 reports robustness results for alternative sorting methodologies, and accounting for transaction costs. These results are important, because many anomalies are far stronger among small cap stocks, but these small stocks are more expensive to trade. Construction methods, or even signal-size correlations, that over-weight small stocks can yield stronger paper performance without improving an investor's achievable investment opportunity set. Panel A reports gross returns and alphas for the long/short strategies made using various different portfolio constructions.

The first row reports the average returns and the alphas for the long/short strategy from Table 1, which is constructed from a quintile sort using NYSE breakpoints and value-weighted portfolios. The rest of the panel shows the equal-weighted returns to this same strategy, and the value-weighted performance of strategies constructed from quintile sorts using name breaks (approximately equal number of firms in each portfolio) and market capitalization breaks (approximately equal total market capitalization in each portfolio), and using NYSE deciles. The average return is lowest for the quintile sort using NYSE breakpoints and equal-weighted portfolios, and equals 22 bps/month with a t-statistics of 2.48. Out of the twenty-five alphas reported in Panel A, the t-statistics for fifteen exceed two, and for nine exceed three.

Panel B reports for these same strategies the average monthly net returns and the generalized net alphas of [Novy-Marx and Velikov \(2016\)](#). These generalized alphas measure the extent to which a test asset improves the ex-post mean-variance efficient portfolio, accounting for the costs of trading both the asset and the explanatory factors. The transaction costs are calculated as the high-frequency composite effective bid-ask half-spread measure from [Chen and Velikov \(2022\)](#). The net average returns reported in the first column range between 12-36bps/month. The lowest return, (12 bps/month), is achieved from the quintile sort using NYSE breakpoints and equal-weighted portfolios, and has an associated t-statistic of 1.28. Out of the twenty-five construction-methodology-factor-model pairs reported in Panel B, the EOB trading strategy improves the achievable mean-variance efficient frontier spanned by the factor models in twenty cases, and significantly expands the achievable frontier in twelve cases.

Table 3 provides direct tests for the role size plays in the EOB strategy performance. Panel A reports the average returns for the twenty-five portfolios constructed from a conditional double sort on size and EOB, as well as average returns and alphas for long/short trading EOB strategies within each size quintile. Panel B reports

the average number of stocks and the average firm size for the twenty-five portfolios. Among the largest stocks (those with market capitalization greater than the 80th NYSE percentile), the EOB strategy achieves an average return of 21 bps/month with a t-statistic of 1.61. Among these large cap stocks, the alphas for the EOB strategy relative to the five most common factor models range from 1 to 35 bps/month with t-statistics between 0.07 and 3.25.

5 How does EOB perform relative to the zoo?

Figure 2 puts the performance of EOB in context, showing the long/short strategy performance relative to other strategies in the “factor zoo.” It shows Sharpe ratio histograms, both for gross and net returns (Panel A and B, respectively), for 212 documented anomalies in the zoo.¹ The vertical red line shows where the Sharpe ratio for the EOB strategy falls in the distribution. The EOB strategy’s gross (net) Sharpe ratio of 0.32 (0.31) is greater than 68% (91%) of anomaly Sharpe ratios, respectively.

Figure 3 plots the growth of a \$1 invested in these same 212 anomaly trading strategies (gray lines), and compares those with the growth of a \$1 invested in the EOB strategy (red line).² Ignoring trading costs, a \$1 invested in the EOB strategy would have yielded \$4.74 which ranks the EOB strategy in the top 6% across the 212 anomalies. Accounting for trading costs, a \$1 invested in the EOB strategy would have yielded \$4.17 which ranks the EOB strategy in the top 2% across the 212 anomalies.

Figure 4 plots percentile ranks for the 212 anomaly trading strategies in terms

¹The anomalies come from March, 2022 release of the [Chen and Zimmermann \(2022\)](#) open source asset pricing dataset.

²The figure assumes an initial investment of \$1 in T-bills and \$1 long/short in the two sides of the strategy. Returns are compounded each month, assuming, as in [Detzel et al. \(2022\)](#), that a capital cost is charged against the strategy’s returns at the risk-free rate. This excess return corresponds more closely to the strategy’s economic profitability.

of gross and [Novy-Marx and Velikov \(2016\)](#) net generalized alphas with respect to the CAPM, and the Fama-French three-, four-, five-, and six-factor models from [Table 1](#), and indicates the ranking of the EOB relative to those. Panel A shows that the EOB strategy gross alphas fall between the 25 and 83 percentiles across the five factor models. Panel B shows that, accounting for trading costs, a large fraction of anomalies have not improved the investment opportunity set of an investor with access to the factor models over the 196306 to 202406 sample. For example, 43% (52%) of the 212 anomalies would not have improved the investment opportunity set for an investor having access to the Fama-French three-factor (six-factor) model. The EOB strategy has a positive net generalized alpha for five out of the five factor models. In these cases EOB ranks between the 52 and 93 percentiles in terms of how much it could have expanded the achievable investment frontier.

6 Does EOB add relative to related anomalies?

With so many anomalies, it is possible that any proposed, new cross-sectional predictor is just capturing some combination of known predictors. It is consequently natural to investigate to what extent the proposed predictor adds additional predictive power beyond the most closely related anomalies. Closely related anomalies are more likely to be formed on the basis of signals with higher absolute correlations. [Figure 5](#) plots a name histogram of the correlations of EOB with 210 filtered anomaly signals.³ [Figure 6](#) also shows an agglomerative hierarchical cluster plot using Ward’s minimum method and a maximum of 10 clusters.

A closely related anomaly is also more likely to price EOB or at least to weaken the power EOB has predicting the cross-section of returns. [Figure 7](#) plots histograms

³When performing tests at the underlying signal level (e.g., the correlations plotted in [Figure 5](#)), we filter the 212 anomalies to avoid small sample issues. For each anomaly, we calculate the common stock observations in an average month for which both the anomaly and the test signal are available. In the filtered anomaly set, we drop anomalies with fewer than 100 common stock observations in an average month.

of t-statistics for predictability tests of EOB conditioning on each of the 210 filtered anomaly signals one at a time. Panel A reports t-statistics on β_{EOB} from Fama-MacBeth regressions of the form $r_{i,t} = \alpha + \beta_{EOB}EOB_{i,t} + \beta_X X_{i,t} + \epsilon_{i,t}$, where X stands for one of the 210 filtered anomaly signals at a time. Panel B plots t-statistics on α from spanning tests of the form: $r_{EOB,t} = \alpha + \beta r_{X,t} + \epsilon_t$, where $r_{X,t}$ stands for the returns to one of the 210 filtered anomaly trading strategies at a time. The strategies employed in the spanning tests are constructed using quintile sorts, value-weighting, and NYSE breakpoints. Panel C plots t-statistics on the average returns to strategies constructed by conditional double sorts. In each month, we sort stocks into quintiles based one of the 210 filtered anomaly signals. Then, within each quintile, we sort stocks into quintiles based on EOB. Stocks are finally grouped into five EOB portfolios by combining stocks within each anomaly sorting portfolio. The panel plots the t-statistics on the average returns of these conditional double-sorted EOB trading strategies conditioned on each of the 210 filtered anomalies.

Table 4 reports Fama-MacBeth cross-sectional regressions of returns on EOB and the six anomalies most closely-related to it. The six most-closely related anomalies are picked as those with the highest combined rank where the ranks are based on the absolute value of the Spearman correlations in Panel B of Figure 5 and the R^2 from the spanning tests in Figure 7, Panel B. Controlling for each of these signals at a time, the t-statistics on the EOB signal in these Fama-MacBeth regressions exceed 1.12, with the minimum t-statistic occurring when controlling for Predicted Analyst forecast error. Controlling for all six closely related anomalies, the t-statistic on EOB is 1.27.

Similarly, Table 5 reports results from spanning tests that regress returns to the EOB strategy onto the returns of the six most closely-related anomalies and the six Fama-French factors. Controlling for the six most-closely related anomalies individually, the EOB strategy earns alphas that range from 25-39bps/month. The

minimum t-statistic on these alphas controlling for one anomaly at a time is 2.88, which is achieved when controlling for Predicted Analyst forecast error. Controlling for all six closely-related anomalies and the six Fama-French factors simultaneously, the EOB trading strategy achieves an alpha of 29bps/month with a t-statistic of 3.03.

7 Does EOB add relative to the whole zoo?

Finally, we can ask how much adding EOB to the entire factor zoo could improve investment performance. Figure 8 plots the growth of \$1 invested in trading strategies that combine multiple anomalies following [Chen and Velikov \(2022\)](#). The combinations use either the 154 anomalies from the zoo that satisfy our inclusion criteria (blue lines) or these 154 anomalies augmented with the EOB signal.⁴ We consider one different methods for combining signals.

Panel A shows results using “Average rank” as the combination method. This method sorts stocks on the basis of forecast excess returns, where these are calculated on the basis of their average cross-sectional percentile rank across return predictors, and the predictors are all signed so that higher ranks are associated with higher average returns. For this method, \$1 investment in the 154-anomaly combination strategy grows to \$2560.30, while \$1 investment in the combination strategy that includes EOB grows to \$3744.48.

8 Conclusion

This study provides compelling evidence that Equity Operational Burden (EOB) represents a robust and economically significant predictor of cross-sectional stock re-

⁴We filter the 207 [Chen and Zimmermann \(2022\)](#) anomalies and require for each anomaly the average month to have at least 40% of the cross-sectional observations available for market capitalization on CRSP in the period for which EOB is available.

turns. Our analysis, conducted using the rigorous protocol established by Novy-Marx and Velikov (2023), demonstrates that EOB-based trading strategies generate substantial risk-adjusted returns that persist even after controlling for well-established risk factors and related anomalies.

The key findings reveal that a value-weighted long/short strategy based on EOB delivers an annualized Sharpe ratio of 0.32 gross (0.31 net), indicating strong performance even after accounting for transaction costs. More importantly, the strategy generates statistically significant abnormal returns of 36 basis points per month (t -statistic = 3.97) relative to the Fama-French five-factor model augmented with momentum. The persistence of a 29 basis point monthly alpha (t -statistic = 3.03) after controlling for six additional closely related factors—including market microstructure variables such as volume-to-market equity and zero-trading days, as well as tail risk measures—underscores the unique information content embedded in the EOB signal.

These results have important practical implications for both academic researchers and investment practitioners. The economic magnitude and statistical robustness of the EOB effect suggest that operational efficiency metrics capture a distinct dimension of firm characteristics that is not fully explained by existing factor models. For portfolio managers, the relatively modest degradation from gross to net returns (36 to 29 basis points) indicates that the strategy remains viable after implementation costs, particularly for institutional investors with access to efficient execution platforms.

However, several limitations warrant consideration. First, while our analysis controls for numerous known factors, the possibility of data mining concerns cannot be entirely eliminated despite following best practices in empirical asset pricing research. Second, the study period and market conditions examined may not fully capture the strategy’s performance across different market regimes or during periods

of extreme market stress. Third, the actual implementation of EOB-based strategies may face additional practical challenges, including capacity constraints and market impact costs at scale, which are not fully captured in our net return calculations.

Future research should explore several promising avenues. First, investigating the economic mechanisms underlying the EOB effect would enhance our understanding of why operational burden predicts returns. Potential explanations might include investor inattention to operational efficiency metrics, limits to arbitrage, or systematic risk exposures not captured by standard factor models. Second, examining the international evidence for EOB across different markets would test the generalizability of our findings and potentially uncover market-specific institutional features that amplify or attenuate the effect. Third, analyzing the time-variation in the EOB premium and its relationship to macroeconomic conditions or market sentiment could provide insights into when the strategy performs best. Finally, combining EOB with machine learning techniques or other alternative data sources might yield enhanced prediction models that better capture the complex relationships between operational characteristics and expected returns.

In conclusion, this paper establishes Equity Operational Burden as a robust and economically meaningful predictor of stock returns that warrants inclusion in the toolkit of both researchers and practitioners interested in understanding and exploiting cross-sectional return predictability.

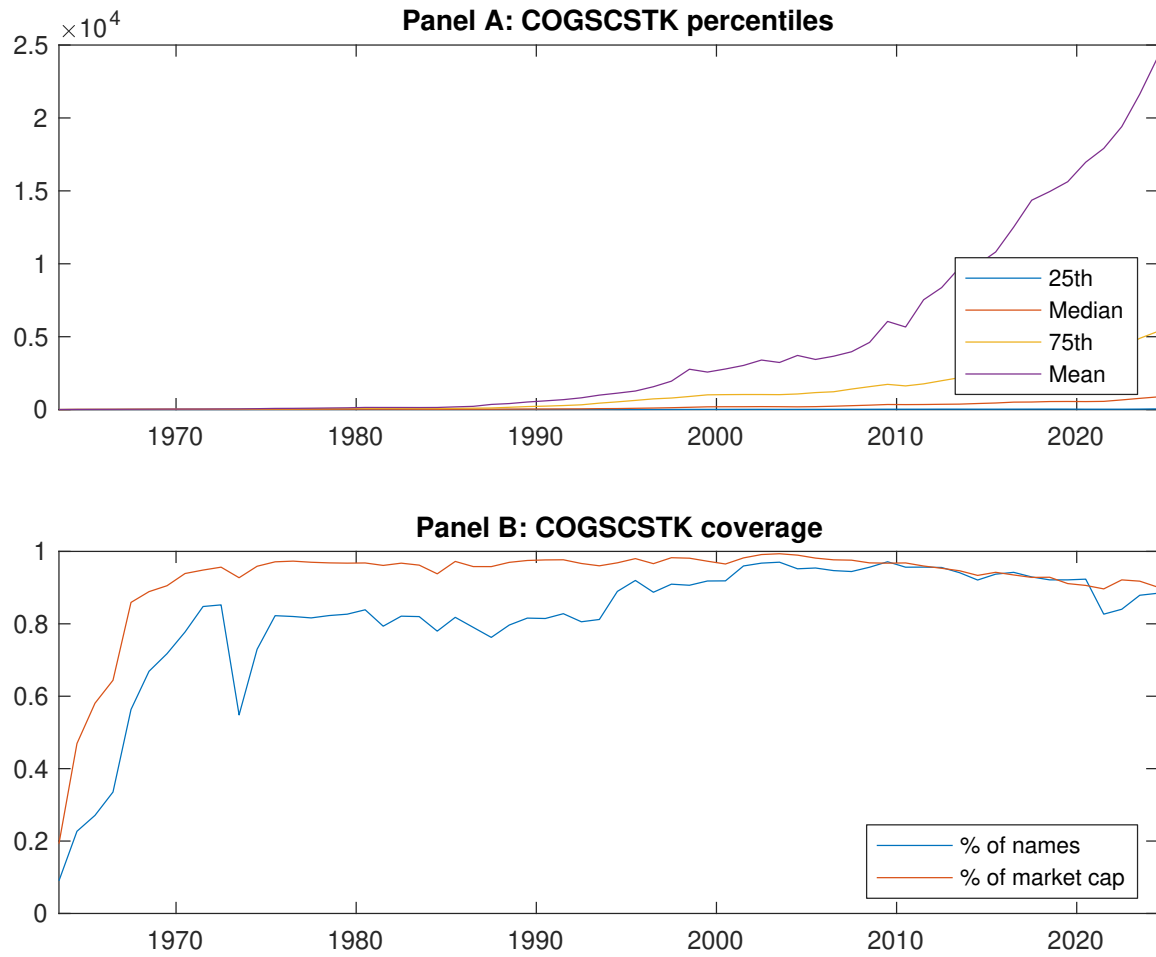


Figure 1: Times series of EOB percentiles and coverage.
This figure plots descriptive statistics for EOB. Panel A shows cross-sectional percentiles of EOB over the sample. Panel B plots the monthly coverage of EOB relative to the universe of CRSP stocks with available market capitalizations.

Table 1: Basic sort: VW, quintile, NYSE-breaks

This table reports average excess returns and alphas for portfolios sorted on EOB. At the end of each month, we sort stocks into five portfolios based on their signal using NYSE breakpoints. Panel A reports average value-weighted quintile portfolio (L,2,3,4,H) returns in excess of the risk-free rate, the long-short extreme quintile portfolio (H-L) return, and alphas with respect to the CAPM, [Fama and French \(1993\)](#) three-factor model, [Fama and French \(1993\)](#) three-factor model augmented with the [Carhart \(1997\)](#) momentum factor, [Fama and French \(2015\)](#) five-factor model, and the [Fama and French \(2015\)](#) five-factor model augmented with the [Carhart \(1997\)](#) momentum factor following [Fama and French \(2018\)](#). Panel B reports the factor loadings for the quintile portfolios and long-short extreme quintile portfolio in the [Fama and French \(2015\)](#) five-factor model. Panel C reports the average number of stocks and market capitalization of each portfolio. T-statistics are in brackets. The sample period is 196306 to 202406.

Panel A: Excess returns and alphas on EOB-sorted portfolios						
	(L)	(2)	(3)	(4)	(H)	(H-L)
r^e	0.48 [3.31]	0.57 [3.51]	0.52 [3.05]	0.62 [3.16]	0.77 [3.91]	0.29 [2.52]
α_{CAPM}	0.00 [0.06]	0.01 [0.32]	-0.07 [-1.79]	-0.06 [-1.12]	0.10 [1.70]	0.10 [0.96]
α_{FF3}	-0.06 [-1.36]	-0.03 [-0.75]	-0.06 [-1.51]	-0.06 [-1.17]	0.16 [2.85]	0.22 [2.50]
α_{FF4}	-0.07 [-1.50]	-0.06 [-1.27]	-0.05 [-1.17]	0.01 [0.28]	0.16 [2.87]	0.23 [2.59]
α_{FF5}	-0.15 [-3.18]	-0.12 [-2.86]	-0.06 [-1.42]	0.02 [0.39]	0.21 [3.76]	0.36 [4.04]
α_{FF6}	-0.15 [-3.14]	-0.13 [-3.11]	-0.05 [-1.13]	0.07 [1.51]	0.21 [3.67]	0.36 [3.97]
Panel B: Fama and French (2018) 6-factor model loadings for EOB-sorted portfolios						
β_{MKT}	0.89 [77.92]	1.00 [99.01]	1.01 [100.49]	1.09 [94.06]	1.06 [78.37]	0.18 [8.16]
β_{SMB}	-0.11 [-6.96]	-0.01 [-0.88]	-0.01 [-0.91]	0.05 [2.79]	0.13 [6.62]	0.24 [7.84]
β_{HML}	0.16 [7.27]	0.07 [3.61]	-0.04 [-2.17]	0.05 [2.09]	-0.11 [-4.23]	-0.27 [-6.50]
β_{RMW}	0.16 [7.31]	0.16 [7.84]	-0.01 [-0.63]	-0.09 [-3.98]	-0.05 [-1.99]	-0.22 [-5.11]
β_{CMA}	0.14 [4.37]	0.16 [5.72]	0.02 [0.73]	-0.19 [-5.70]	-0.17 [-4.51]	-0.31 [-5.14]
β_{UMD}	0.00 [0.01]	0.02 [1.73]	-0.02 [-1.62]	-0.08 [-6.70]	0.00 [0.17]	0.00 [0.10]
Panel C: Average number of firms (n) and market capitalization (me)						
n	860	629	720	734	783	
me (\$10 ⁶)	2287	1948	1705	1576	2319	

Table 2: Robustness to sorting methodology & trading costs

This table evaluates the robustness of the choices made in the EOB strategy construction methodology. In each panel, the first row shows results from a quintile, value-weighted sort using NYSE break points as employed in Table 1. Each of the subsequent rows deviates in one of the three choices at a time, and the choices are specified in the first three columns. For each strategy construction methodology, the table reports average excess returns and alphas with respect to the CAPM, Fama and French (1993) three-factor model, Fama and French (1993) three-factor model augmented with the Carhart (1997) momentum factor, Fama and French (2015) five-factor model, and the Fama and French (2015) five-factor model augmented with the Carhart (1997) momentum factor following Fama and French (2018). Panel A reports average returns and alphas with no adjustment for trading costs. Panel B reports net average returns and Novy-Marx and Velikov (2016) generalized alphas as prescribed by Detzel et al. (2022). T-statistics are in brackets. The sample period is 196306 to 202406.

Panel A: Gross Returns and Alphas								
Portfolios	Breaks	Weights	r^e	α_{CAPM}	α_{FF3}	α_{FF4}	α_{FF5}	α_{FF6}
Quintile	NYSE	VW	0.29 [2.52]	0.10 [0.96]	0.22 [2.50]	0.23 [2.59]	0.36 [4.04]	0.36 [3.97]
Quintile	NYSE	EW	0.22 [2.48]	0.05 [0.62]	0.04 [0.57]	0.11 [1.65]	0.02 [0.27]	0.08 [1.22]
Quintile	Name	VW	0.27 [2.21]	0.06 [0.53]	0.18 [2.03]	0.21 [2.37]	0.31 [3.41]	0.32 [3.55]
Quintile	Cap	VW	0.23 [1.92]	0.03 [0.27]	0.17 [1.82]	0.21 [2.25]	0.30 [3.22]	0.33 [3.45]
Decile	NYSE	VW	0.38 [2.59]	0.14 [1.06]	0.33 [3.08]	0.32 [2.92]	0.44 [4.04]	0.42 [3.80]
Panel B: Net Returns and Novy-Marx and Velikov (2016) generalized alphas								
Portfolios	Breaks	Weights	r_{net}^e	α_{CAPM}^*	α_{FF3}^*	α_{FF4}^*	α_{FF5}^*	α_{FF6}^*
Quintile	NYSE	VW	0.28 [2.40]	0.08 [0.78]	0.19 [2.16]	0.20 [2.23]	0.29 [3.30]	0.29 [3.27]
Quintile	NYSE	EW	0.12 [1.28]					
Quintile	Name	VW	0.25 [2.08]	0.04 [0.33]	0.15 [1.67]	0.17 [1.88]	0.23 [2.61]	0.24 [2.71]
Quintile	Cap	VW	0.22 [1.81]	0.01 [0.14]	0.14 [1.53]	0.17 [1.79]	0.23 [2.51]	0.25 [2.67]
Decile	NYSE	VW	0.36 [2.47]	0.12 [0.90]	0.29 [2.69]	0.29 [2.64]	0.36 [3.32]	0.35 [3.22]

Table 3: Conditional sort on size and EOB

This table presents results for conditional double sorts on size and EOB. In each month, stocks are first sorted into quintiles based on size using NYSE breakpoints. Then, within each size quintile, stocks are further sorted based on EOB. Finally, they are grouped into twenty-five portfolios based on the intersection of the two sorts. Panel A presents the average returns to the 25 portfolios, as well as strategies that go long stocks with high EOB and short stocks with low EOB. Panel B documents the average number of firms and the average firm size for each portfolio. The sample period is 1963:06 to 2024:06.

Panel A: portfolio average returns and time-series regression results												
Size quintiles	EOB Quintiles					EOB Strategies						
	(L)	(2)	(3)	(4)	(H)	r^e	α_{CAPM}	α_{FF3}	α_{FF4}	α_{FF5}	α_{FF6}	
	(1)	0.48 [2.18]	0.71 [3.10]	0.82 [3.27]	0.74 [2.97]	0.93 [3.41]	0.44 [3.70]	0.29 [2.60]	0.25 [2.23]	0.31 [2.73]	0.19 [1.64]	0.25 [2.16]
	(2)	0.63 [3.20]	0.80 [3.73]	0.83 [3.48]	0.78 [3.23]	0.84 [3.45]	0.21 [1.96]	0.06 [0.58]	0.12 [1.29]	0.10 [1.05]	0.09 [0.93]	0.07 [0.77]
	(3)	0.53 [3.12]	0.68 [3.38]	0.82 [3.85]	0.81 [3.61]	0.84 [3.62]	0.31 [2.56]	0.11 [0.98]	0.18 [1.94]	0.16 [1.72]	0.15 [1.59]	0.14 [1.44]
	(4)	0.57 [3.48]	0.61 [3.33]	0.69 [3.43]	0.73 [3.56]	0.80 [3.73]	0.23 [2.02]	0.04 [0.44]	0.16 [1.87]	0.15 [1.68]	0.17 [1.98]	0.16 [1.80]
	(5)	0.48 [3.35]	0.48 [3.01]	0.47 [2.81]	0.56 [3.08]	0.69 [3.62]	0.21 [1.61]	0.01 [0.07]	0.15 [1.42]	0.20 [1.87]	0.32 [3.02]	0.35 [3.25]
Panel B: Portfolio average number of firms and market capitalization												
Size quintiles	EOB Quintiles					EOB Quintiles						
	Average n					Average market capitalization (\$10 ⁶)						
	(L)	(2)	(3)	(4)	(H)	(L)	(2)	(3)	(4)	(H)		
	(1)	418	420	419	418	417	32	33	33	36	40	
	(2)	116	116	116	115	116	59	59	59	60	59	
	(3)	83	82	82	82	82	102	101	101	101	101	
	(4)	68	68	68	68	68	216	214	212	212	206	
(5)	61	61	61	61	61	1564	1368	1405	1383	2078		

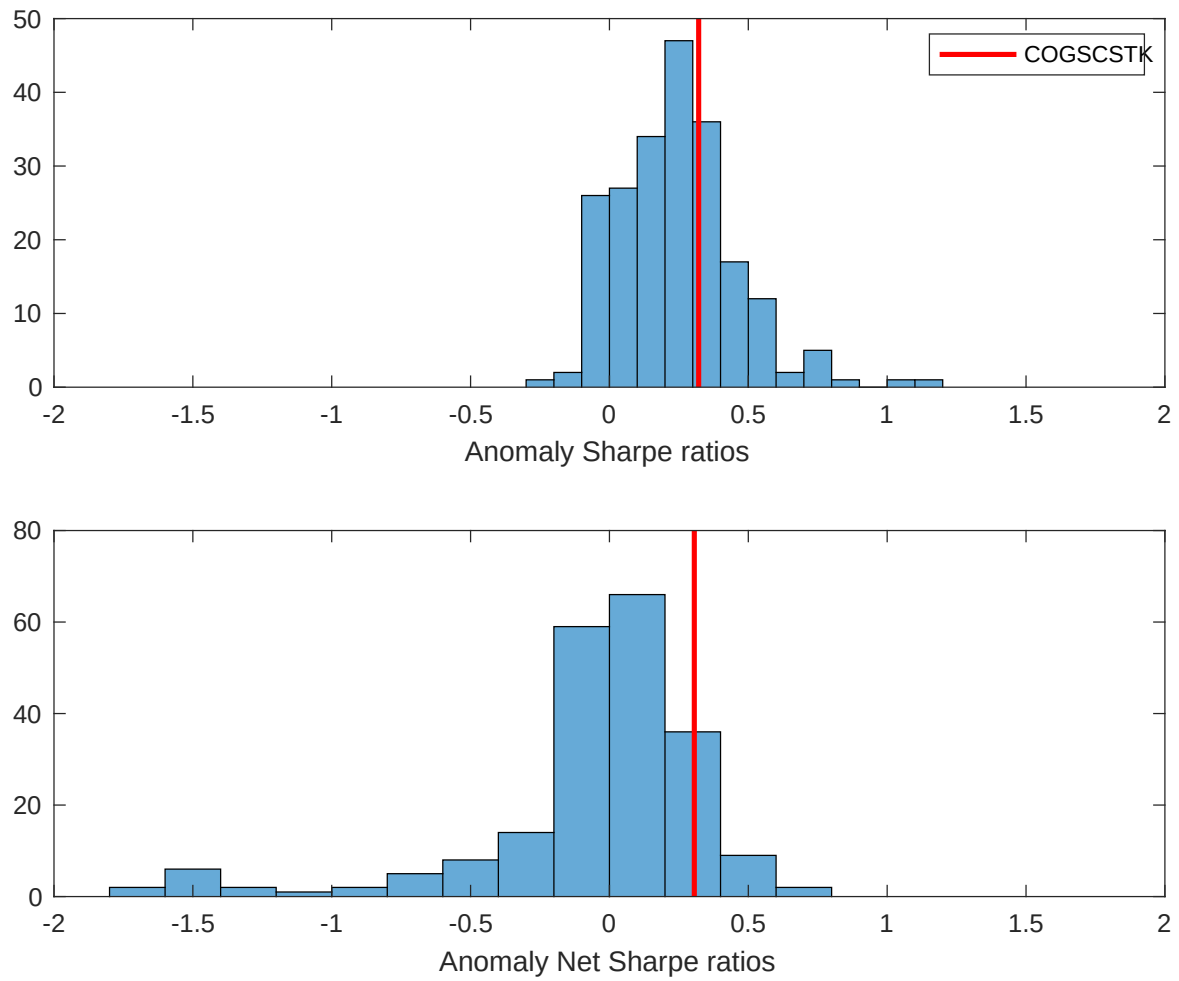


Figure 2: Distribution of Sharpe ratios.

This figure plots a histogram of Sharpe ratios for 212 anomalies, and compares the Sharpe ratio of the EOB with them (red vertical line). Panel A plots results for gross Sharpe ratios. Panel B plots results for net Sharpe ratios.

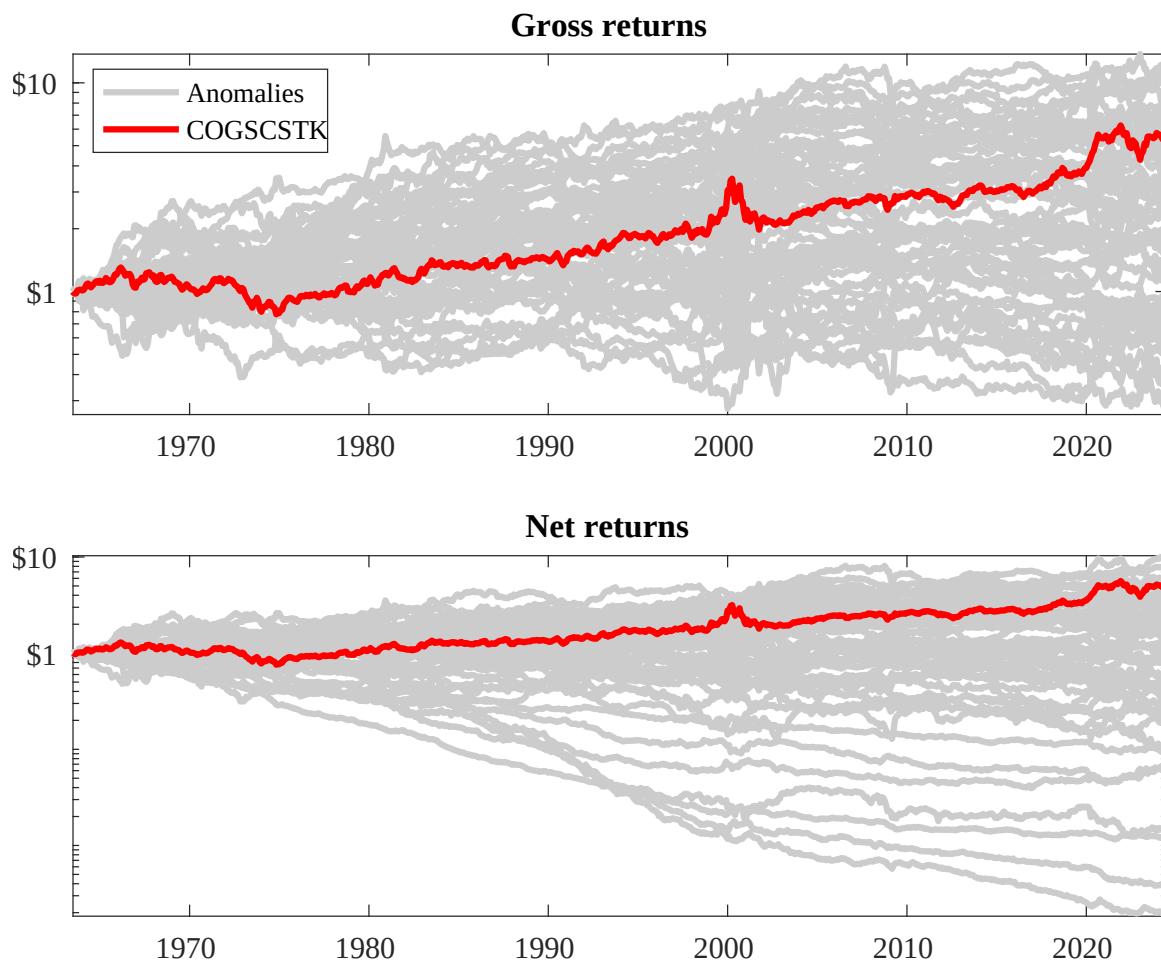
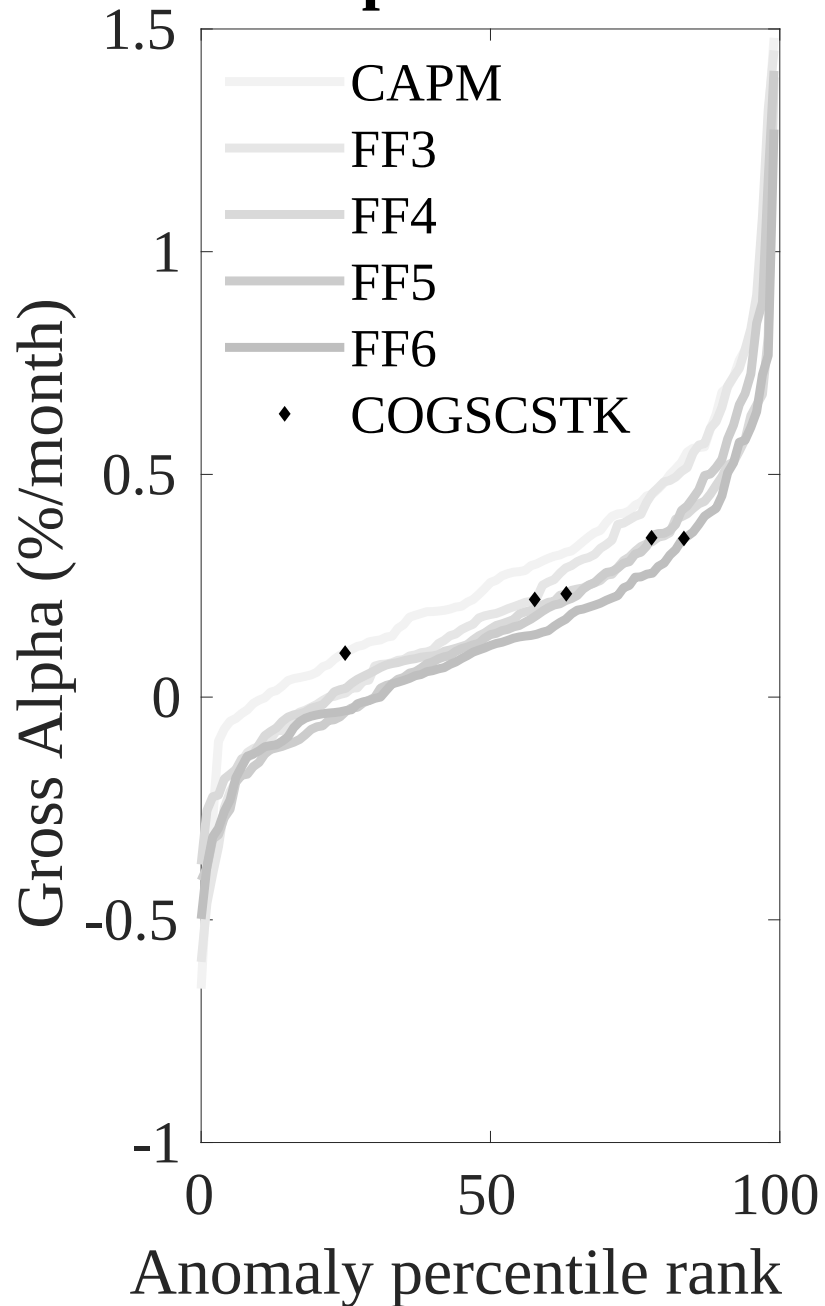


Figure 3: Dollar invested.

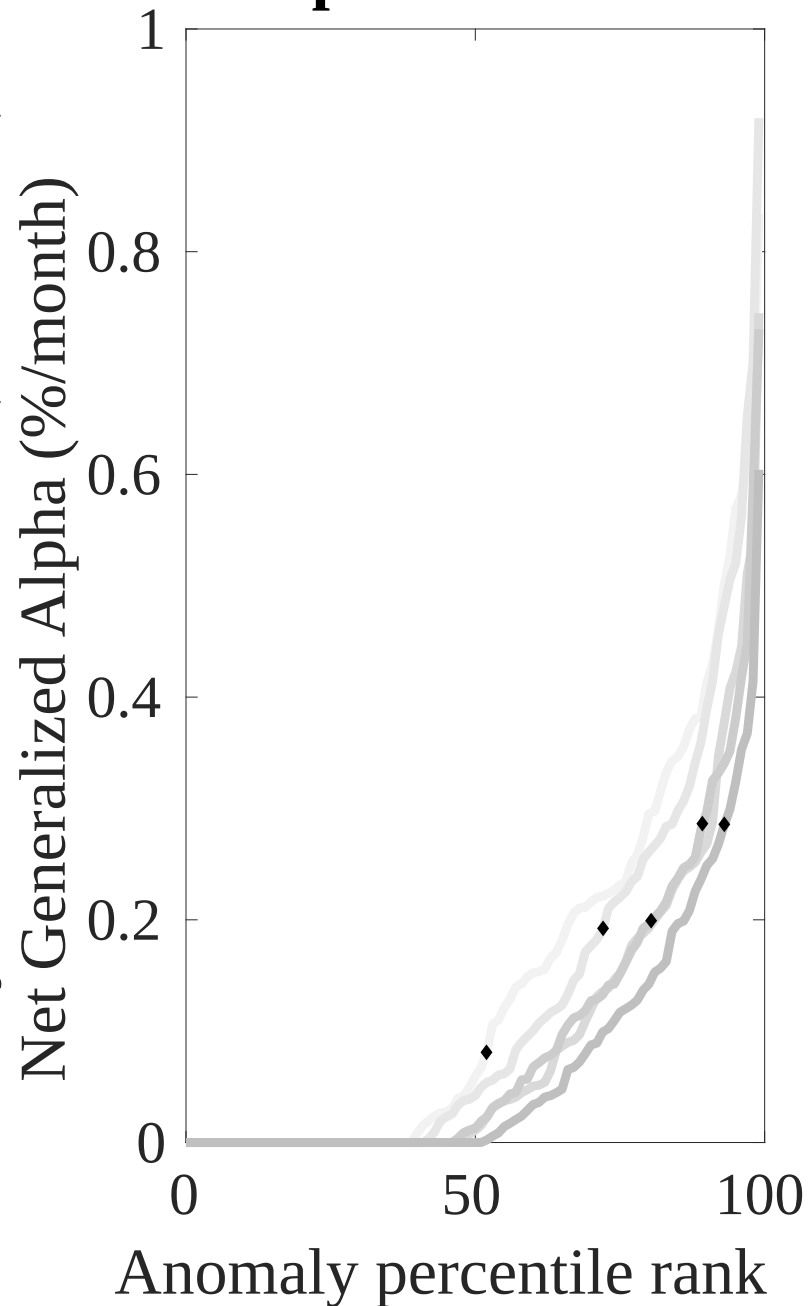
This figure plots the growth of a \$1 invested in 212 anomaly trading strategies (gray lines), and compares those with the EOB trading strategy (red line). The strategies are constructed using value-weighted quintile sorts using NYSE breakpoints. Panel A plots results for gross strategy returns. Panel B plots results for net strategy returns.

Gross Alpha Percentiles



Novy-Marx and Velikov (RFS, 2016)

Net Alpha Percentiles



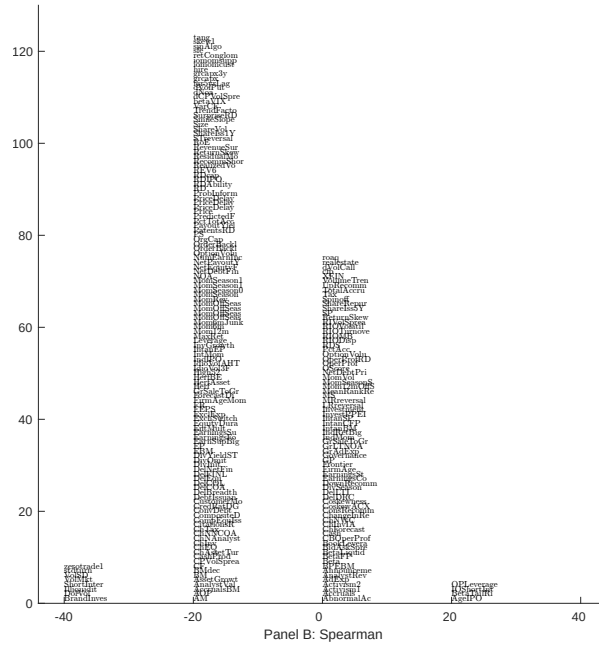
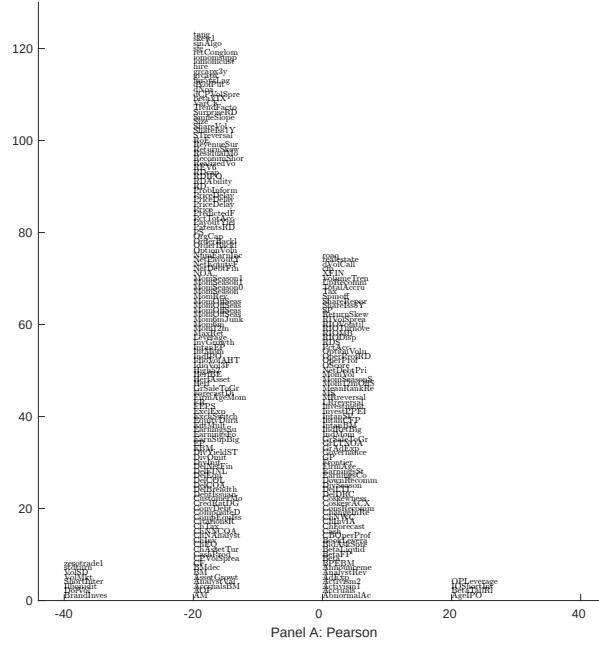


Figure 5: Distribution of correlations.
This figure plots a name histogram of correlations of 210 filtered anomaly signals with EOB. The correlations are pooled. Panel A plots Pearson correlations, while Panel B plots Spearman rank correlations.

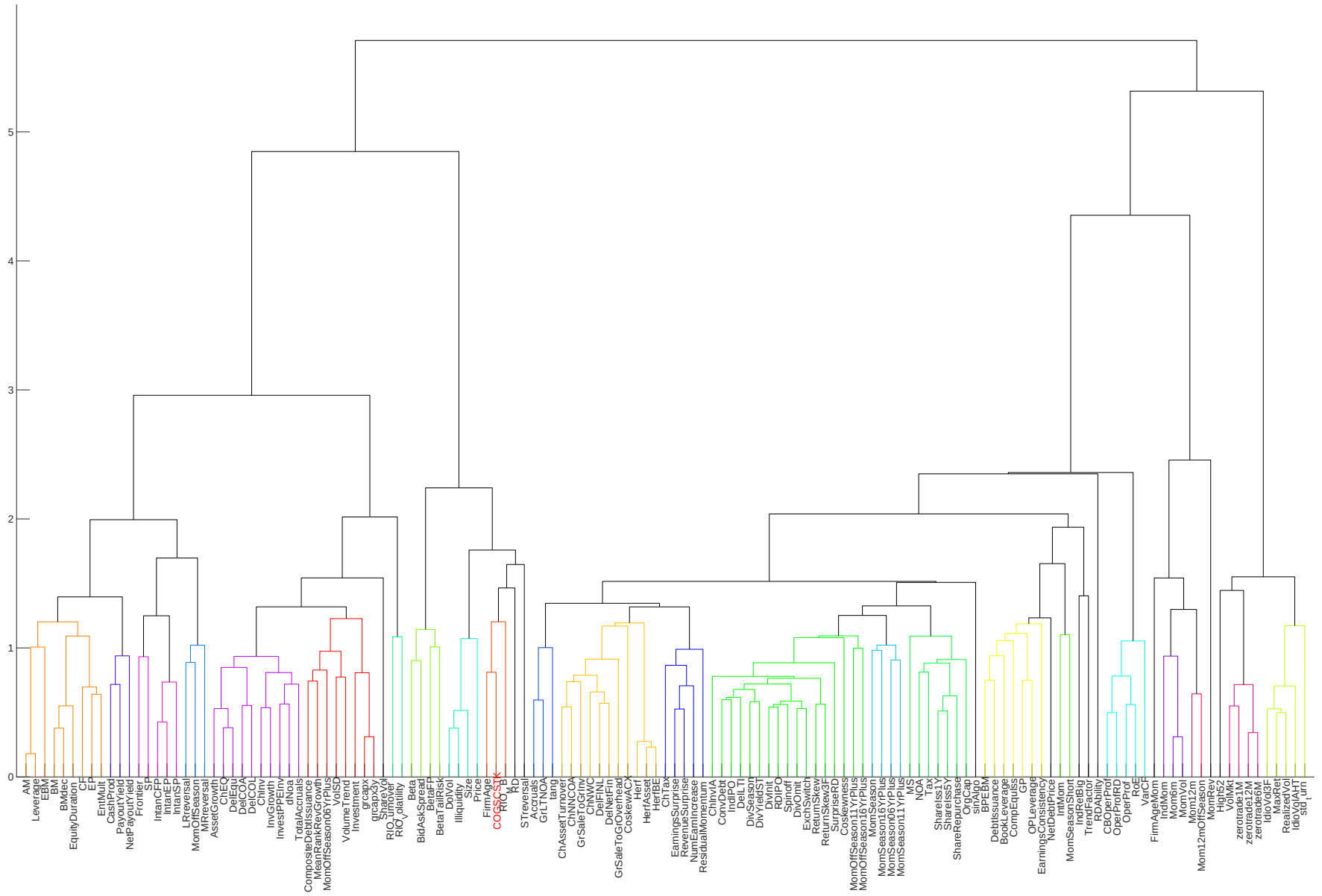


Figure 6: Agglomerative hierarchical cluster plot

This figure plots an agglomerative hierarchical cluster plot using Ward's minimum method and a maximum of 10 clusters.

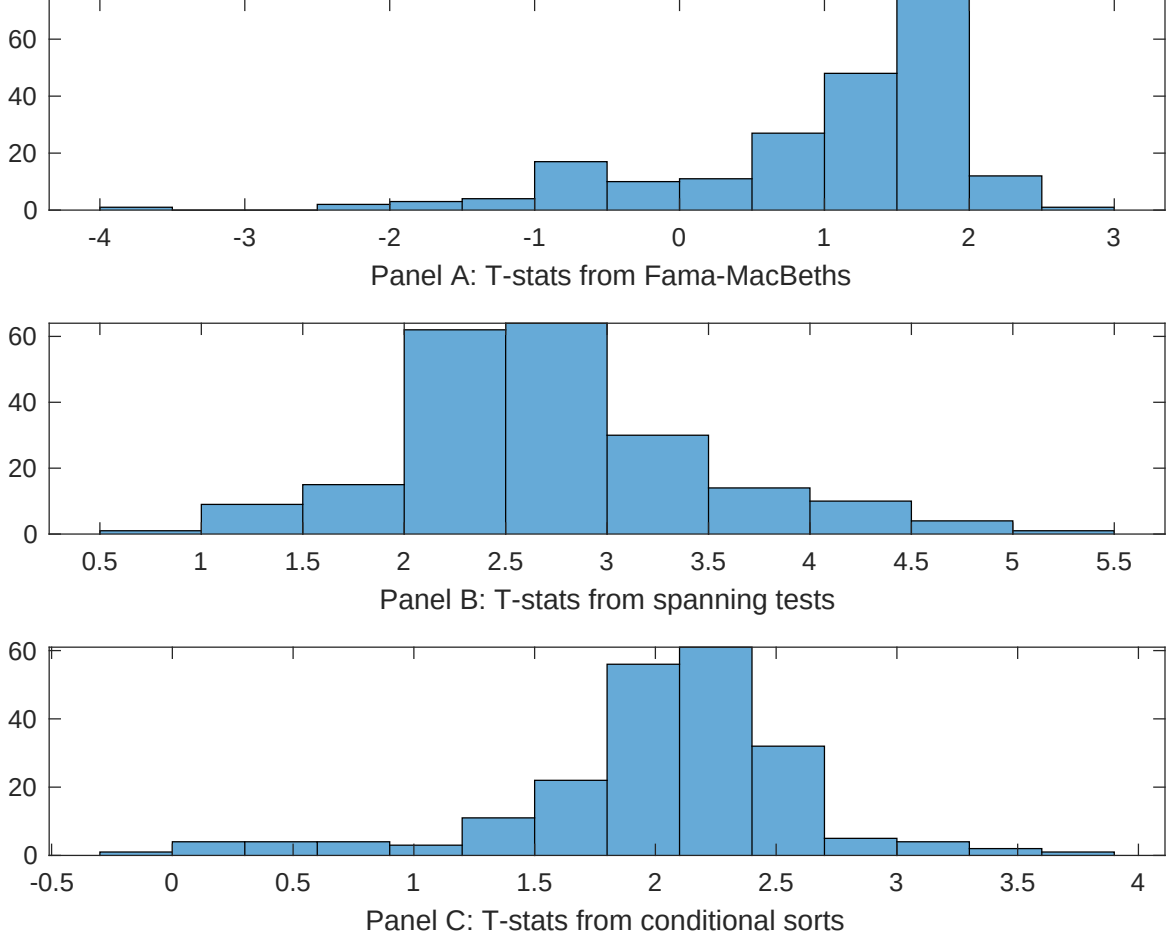


Figure 7: Distribution of t-stats on conditioning strategies

This figure plots histograms of t-statistics for predictability tests of EOB conditioning on each of the 210 filtered anomaly signals one at a time. Panel A reports t-statistics on β_{EOB} from Fama-MacBeth regressions of the form $r_{i,t} = \alpha + \beta_{EOB}EOB_{i,t} + \beta_X X_{i,t} + \epsilon_{i,t}$, where X stands for one of the 210 filtered anomaly signals at a time. Panel B plots t-statistics on α from spanning tests of the form: $r_{EOB,t} = \alpha + \beta r_{X,t} + \epsilon_t$, where $r_{X,t}$ stands for the returns to one of the 210 filtered anomaly trading strategies at a time. The strategies employed in the spanning tests are constructed using quintile sorts, value-weighting, and NYSE breakpoints. Panel C plots t-statistics on the average returns to strategies constructed by conditional double sorts. In each month, we sort stocks into quintiles based one of the 210 filtered anomaly signals at a time. Then, within each quintile, we sort stocks into quintiles based on EOB. Stocks are finally grouped into five EOB portfolios by combining stocks within each anomaly sorting portfolio. The panel plots the t-statistics on the average returns of these conditional double-sorted EOB trading strategies conditioned on each of the 210 filtered anomalies.

Table 4: Fama-MacBeths controlling for most closely related anomalies

This table presents Fama-MacBeth results of returns on EOB. and the six most closely related anomalies. The regressions take the following form: $r_{i,t} = \alpha + \beta_{EOB}EOB_{i,t} + \sum_{k=1}^s \beta_{X_k} X_{i,t}^k + \epsilon_{i,t}$. The six most closely related anomalies, X , are Volume to market equity, Days with zero trades, Days with zero trades, Days with zero trades, Tail risk beta, Predicted Analyst forecast error. These anomalies were picked as those with the highest combined rank where the ranks are based on the absolute value of the Spearman correlations in Panel B of Figure 5 and the R^2 from the spanning tests in Figure 7, Panel B. The sample period is 196306 to 202406.

Intercept	0.13 [7.36]	0.11 [4.76]	0.10 [4.61]	0.10 [4.64]	0.97 [6.05]	0.12 [5.33]	0.11 [6.28]
EOB	0.60 [1.40]	0.85 [1.74]	0.85 [1.76]	0.87 [1.79]	0.42 [1.17]	0.28 [1.12]	0.30 [1.27]
Anomaly 1	0.20 [1.53]						0.15 [2.47]
Anomaly 2		0.23 [1.38]					0.61 [0.26]
Anomaly 3			0.19 [2.87]				-0.65 [-0.72]
Anomaly 4				0.98 [2.98]			0.39 [0.23]
Anomaly 5					0.51 [3.13]		0.31 [1.70]
Anomaly 6						0.17 [0.65]	0.11 [0.58]
# months	727	727	727	727	727	492	487
$\bar{R}^2(\%)$	2	1	1	1	1	1	0

Table 5: Spanning tests controlling for most closely related anomalies

This table presents spanning tests results of regressing returns to the EOB trading strategy on trading strategies exploiting the six most closely related anomalies. The regressions take the following form: $r_t^{EOB} = \alpha + \sum_{k=1}^6 \beta_{X_k} r_t^{X_k} + \sum_{j=1}^6 \beta_{f_j} r_t^{f_j} + \epsilon_t$, where X_k indicates each of the six most-closely related anomalies and f_j indicates the six factors from the [Fama and French \(2015\)](#) five-factor model augmented with the [Carhart \(1997\)](#) momentum factor. The six most closely related anomalies, X , are Volume to market equity, Days with zero trades, Days with zero trades, Days with zero trades, Tail risk beta, Predicted Analyst forecast error. These anomalies were picked as those with the highest combined rank where the ranks are based on the absolute value of the Spearman correlations in Panel B of Figure 5 and the R^2 from the spanning tests in Figure 7, Panel B. The sample period is 196306 to 202406.

Intercept	0.27 [3.16]	0.32 [3.70]	0.36 [4.15]	0.33 [3.83]	0.25 [2.88]	0.39 [3.87]	0.29 [3.03]
Anomaly 1	-27.60 [-10.54]						-13.40 [-1.97]
Anomaly 2		-23.99 [-9.25]					0.29 [0.04]
Anomaly 3			-22.17 [-8.46]				-1.67 [-0.17]
Anomaly 4				-23.15 [-8.80]			-9.70 [-1.15]
Anomaly 5					22.89 [9.12]		6.44 [1.98]
Anomaly 6						-24.06 [-7.94]	-15.02 [-4.81]
mkt	5.15 [2.20]	7.56 [3.26]	8.19 [3.49]	8.05 [3.46]	8.26 [3.61]	13.44 [5.49]	0.63 [0.23]
smb	8.22 [2.48]	15.80 [5.07]	21.14 [7.03]	23.73 [8.01]	15.50 [4.95]	23.63 [6.47]	11.71 [2.79]
hml	-18.82 [-4.80]	-20.35 [-5.12]	-20.53 [-5.12]	-21.00 [-5.27]	-28.66 [-7.25]	-14.95 [-3.15]	-14.86 [-3.26]
rmw	-6.79 [-1.63]	-6.68 [-1.56]	-7.31 [-1.68]	-7.97 [-1.86]	-13.13 [-3.22]	-9.87 [-2.07]	3.77 [0.78]
cma	-16.48 [-2.81]	-18.72 [-3.15]	-18.90 [-3.14]	-17.58 [-2.92]	-15.35 [-2.53]	-44.38 [-6.80]	-28.06 [-4.31]
umd	6.32 [3.07]	-0.05 [-0.03]	0.63 [0.31]	1.26 [0.62]	3.46 [1.70]	1.37 [0.60]	8.28 [3.14]
# months	728	728	728	728	728	492	488
$\bar{R}^2(\%)$	54	52	51	52	52	61	66

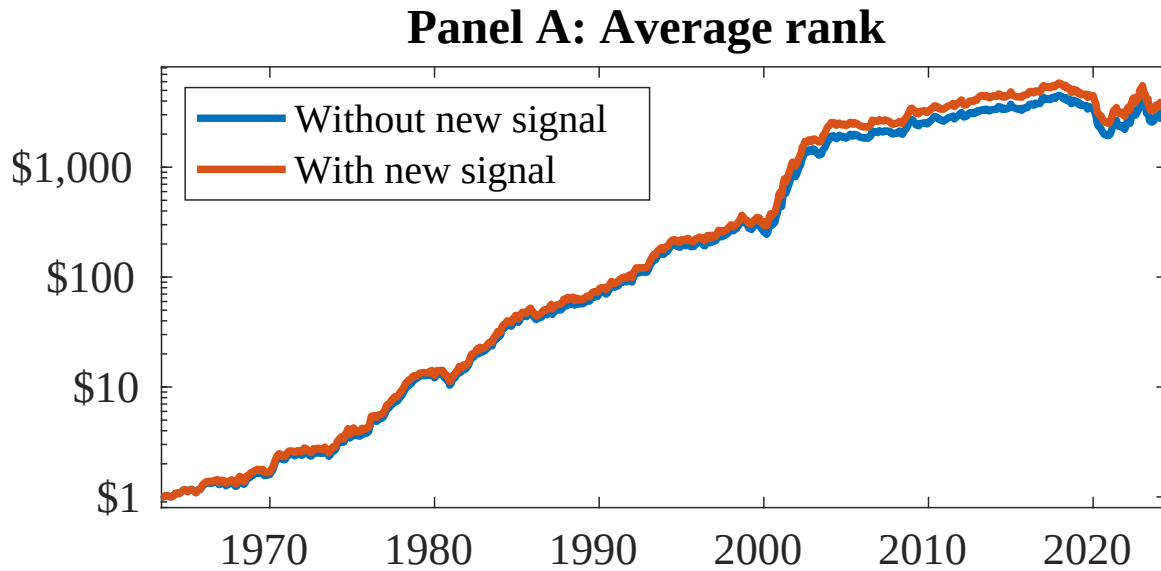


Figure 8: Combination strategy performance

This figure plots the growth of a \$1 invested in trading strategies that combine multiple anomalies following [Chen and Velikov \(2022\)](#). In all panels, the blue solid lines indicate combination trading strategies that utilize 154 anomalies. The red solid lines indicate combination trading strategies that utilize the 154 anomalies as well as EOB. Panel A shows results using "Average rank" as the combination method. See Section 7 for details on the combination methods.

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